

Data Cleaning & Preparation - Change Log

DecodeLabs Industrial Training Kit | Data Analytics - Project 1 | Prepared by Pritesh Raj

1. Project Overview

This report documents every data quality issue identified in the raw e-commerce orders dataset (1200 records, 14 columns) and the exact corrective action applied to resolve it. As stated in the project brief: stakeholders need to know WHAT was changed and WHY.

2. Issues Identified (Before Cleaning)

Issue	Details
Missing values	309 records (25.75%) had a blank 'CouponCode'
Date format	Dates were not consistently in ISO 8601 (YYYY-MM-DD) text form
Numeric precision	'UnitPrice' and 'TotalPrice' had inconsistent decimal places (e.g. 180.5 instead of 180.50)
Duplicate records	Verified for fully duplicated rows and duplicate OrderIDs

3. Change Log (Actions Taken)

Change ID	Description	Impact	Status
CR001	Imputed missing 'CouponCode' values with 'No Coupon'	Preserved 309 records (no rows deleted)	Resolved
CR002	Checked and removed fully duplicated rows	Removed 0 duplicate row(s)	Resolved
CR003	Verified OrderID column for duplicate unique identifiers	Removed 0 duplicate OrderID record(s)	Resolved
CR004	Standardized 'Date' column to ISO 8601 format (YYYY-MM-DD)	Reformatted 1200 record(s)	Resolved
CR005	Standardized 'UnitPrice' and 'TotalPrice' to 2 decimal precision	Reformatted 1200 record(s)	Resolved
CR006	Validated TotalPrice = Quantity x UnitPrice for all records	Found 0 mismatch(es) after rounding correction	Resolved

4. Verification Gate Results

Metric	Result	Target
Duplicate Unique Identifiers (OrderID)	0	0

Incorrectly Formatted Dates	0	0
Remaining Missing/Null Values	0	0
Records Before Cleaning	1200	-
Records After Cleaning	1200	-

5. Tools Used

Python 3, Pandas, and OpenPyXL were used to perform a reproducible, script-based cleaning workflow, as recommended for datasets of this scale (Excel Power Query is best suited for under 1 million rows when a visual workflow is preferred; Python/Pandas was chosen here for an automated and version-controllable pipeline).